

CODING TECHNICAL ASSESSMENT

Position: Full Stack Java Developer

Level: Mid-level

Technologies: Java, HTML, CSS, JavaScript

INSTRUCTIONS:

- This is a coding-focused technical assessment
- Complete all coding exercises and projects
- Submit your code files with clear documentation
- Time limit: 2-3 hours
- Reply to this email with your completed solutions

****Full Stack Java Developer Coding Assessment****

****Time Limit:** 3 hours**

****Instructions:****

1. Create a new directory for the assessment and name it `full-stack-java-developer-assessment`.
2. Create separate subdirectories for each section (e.g., `coding-exercises`, `practical-projects`, `code-review-tasks`).
3. Name each file according to the task number (e.g., `coding-exercise-1.java`, `practical-project-1.html`).
4. Use Java 8 or later for Java-related tasks.
5. Use HTML5, CSS3, and JavaScript for frontend-related tasks.
6. Use MySQL or PostgreSQL for database-related tasks (if applicable).
7. Use a code editor or IDE of your choice.
8. Run and test your code before submitting.

****Submission Guidelines:****

1. Zip the entire directory (`full-stack-java-developer-assessment`) and submit it.
2. Ensure that all files are included and correctly named.

3. Include a brief explanation of your design decisions and problem-solving approaches in a separate file named `README.md`.

****CODING EXERCISES (7 questions)****

1. Algorithm: Find the First Duplicate in an Array

****Problem Statement:****

Given an array of integers, find the first duplicate element.

****Input/Output Examples:****

* Input: `[2, 3, 5, 2, 7, 3, 1]`

* Output: `2`

* Input: `[1, 2, 3, 4, 5]`

* Output: `null`

****Expected Time to Complete:** 15 minutes**

****Evaluation Criteria:****

* Correctness

* Efficiency (time complexity)

* Code quality

****Sample Test Cases:****

* `findFirstDuplicate(new int[] {2, 3, 5, 2, 7, 3, 1})` should return `2`

* `findFirstDuplicate(new int[] {1, 2, 3, 4, 5})` should return `null`

```java

```
public class FirstDuplicateFinder {
 public static Integer findFirstDuplicate(int[] array) {
```

```
 // Implement your solution here
 }
}
...

```

## ### 2. Code Implementation: Implement a Simple Queue

### **\*\*Problem Statement:\*\***

Implement a basic queue data structure in Java.

### **\*\*Input/Output Examples:\*\***

```
* Input: `enqueue(1)`, `enqueue(2)`, `dequeue()`
* Output: `1`
* Input: `enqueue(3)`, `dequeue()`, `dequeue()`
* Output: `3`, `null`

```

### **\*\*Expected Time to Complete:\*\* 20 minutes**

### **\*\*Evaluation Criteria:\*\***

- \* Correctness
- \* Code quality
- \* Efficiency (time complexity)

### **\*\*Sample Test Cases:\*\***

```
* `queue.enqueue(1)`, `queue.enqueue(2)`, `queue.dequeue()` should return `1`
* `queue.enqueue(3)`, `queue.dequeue()` should return `3`

```

```
```java
public class SimpleQueue {
    private Node head;

```

```
private Node tail;

public void enqueue(int value) {
    // Implement your solution here
}

public Integer dequeue() {
    // Implement your solution here
}
}
```

3. Bug Fixing: Fix the Broken Calculator

****Problem Statement:****

A simple calculator program is not working correctly. Fix the bugs.

****Input/Output Examples:****

* Input: `calculate(2, '+', 3)`

* Output: `5`

* Input: `calculate(5, '-', 2)`

* Output: `3`

****Expected Time to Complete:** 15 minutes**

****Evaluation Criteria:****

- * Correctness
- * Code quality
- * Efficiency (time complexity)

****Sample Test Cases:****

* `calculate(2, '+', 3)` should return `5`
* `calculate(5, '-', 2)` should return `3`

```
```java
public class Calculator {
 public static int calculate(int num1, char operator, int num2) {
 // Fix the bugs in this method
 }
}
```
```

4. API Development: Implement a Simple REST API

****Problem Statement:****

Implement a simple REST API to manage books.

****Input/Output Examples:****

* Input: `GET /books`
* Output: `[{"id": 1, "title": "Book 1"}, {"id": 2, "title": "Book 2"}]`
* Input: `POST /books` with `{"title": "New Book"}`
* Output: `{"id": 3, "title": "New Book"}`

****Expected Time to Complete:**** 30 minutes

****Evaluation Criteria:****

- * Correctness
- * Code quality
- * Efficiency (time complexity)

****Sample Test Cases:****

- * `GET /books` should return a list of books
- * `POST /books` with `{ "title": "New Book" }` should create a new book

```
``java
import javax.ws.rs.Consumes;
import javax.ws.rs.GET;
import javax.ws.rs.POST;
import javax.ws.rs.Path;
import javax.ws.rs.Produces;
import javax.ws.rs.core.MediaType;
```

```
@Path("/books")
public class BookResource {
    // Implement your solution here
}
``
```

5. Database Query: Retrieve Books by Author

****Problem Statement:****

Write a SQL query to retrieve books written by a specific author.

****Input/Output Examples:****

* Input: `author = "John Doe"`

* Output: `SELECT \* FROM books WHERE author = "John Doe"`

****Expected Time to Complete:**** 10 minutes

****Evaluation Criteria:****

* Correctness

* Efficiency (query optimization)

****Sample Test Cases:****

* `author = "John Doe"` should return a list of books written by John Doe

```
```sql
```

```
SELECT * FROM books WHERE author = ?
```

```
```
```

6. Code Optimization: Optimize the Slow Algorithm

****Problem Statement:****

Optimize the slow algorithm to find the maximum value in an array.

****Input/Output Examples:****

* Input: `[1, 2, 3, 4, 5]`

* Output: `5`

* Input: `[5, 4, 3, 2, 1]`

* Output: `5`

****Expected Time to Complete:**** 15 minutes

****Evaluation Criteria:****

* Correctness

* Efficiency (time complexity)

****Sample Test Cases:****

* `[1, 2, 3, 4, 5]` should return `5`

* `[5, 4, 3, 2, 1]` should return `5`

```
``java
public class MaxValueFinder {
    public static int findMaxValue(int[] array) {
        // Optimize the algorithm here
    }
}
``
```

7. Code Implementation: Implement a Simple Chat System

****Problem Statement:****

Implement a simple chat system using Java.

****Input/Output Examples:****

* Input: ``send("Hello, World!")``
* Output: ``"Hello, World!"``
* Input: ``send("Hi, there!")``
* Output: ``"Hi, there!"``

****Expected Time to Complete:** 30 minutes**

****Evaluation Criteria:****

- * Correctness
- * Code quality
- * Efficiency (time complexity)

****Sample Test Cases:****

* ``send("Hello, World!")`` should return ``"Hello, World!"``
* ``send("Hi, there!")`` should return ``"Hi, there!"``


```
```java
public class ChatSystem {
 public static String send(String message) {
 // Implement your solution here
 }
}
```
```

****PRACTICAL PROJECTS (2 mini-projects)****

1. Build a Simple Todo List App

****Problem Statement:****

Build a simple todo list app using HTML, CSS, and JavaScript.

****Input/Output Examples:****

* Input: `addTodo("Buy milk")`

* Output: `["Buy milk"]`

* Input: `removeTodo(0)`

* Output: `[]`

****Expected Time to Complete:** 45 minutes**

****Evaluation Criteria:****

- * Correctness
- * Code quality
- * User experience

****Sample Test Cases:****

- * `addTodo("Buy milk")` should add a new todo item
- * `removeTodo(0)` should remove the first todo item

```
```html
<!-- Implement your solution here -->
```
```

2. Build a Simple Weather App

****Problem Statement:****

Build a simple weather app using HTML, CSS, and JavaScript.

****Input/Output Examples:****

- * Input: `getWeather("New York")`
- * Output: `{ "temperature": 22, "humidity": 60 }`
- * Input: `getWeather("London")`
- * Output: `{ "temperature": 18, "humidity": 50 }`

****Expected Time to Complete:**** 45 minutes

****Evaluation Criteria:****

- * Correctness
- * Code quality
- * User experience

****Sample Test Cases:****

- * `getWeather("New York")` should return the weather data for New York
- * `getWeather("London")` should return the weather data for London

```
```html
```

<!-- Implement your solution here -->

``

**\*\*CODE REVIEW TASKS (2 questions)\*\***

### 1. Review the Given Code and Improve It

**\*\*Problem Statement:\*\***

Review the given code and improve it for better performance and readability.

**\*\*Input/Output Examples:\*\***

\* Input: ``improveCode("given_code.java")``

\* Output: ``"improved_code.java"``

**\*\*Expected Time to Complete:\*\***

Generated on: 15/09/2025 14:35:07